Can generative AI models downscale very rare events? A study of the 2020 south of France flash flood.

Pierre Chapel[1], Olivier Boucher[2], Freddy Bouchet[1], Kazem Ardaneh[2], Kishanthan Kingston[2], Redouane Lguensat[2]

1-Laboratoire de Météorologie Dynamique (LMD) - Institut Pierre Simon Laplace (IPSL)

2-Institut Pierre Simon Laplace (IPSL)

Corresponding author: pierre.chapel@lmd.ipsl.fr

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1 Introduction and objectives

Downscaling of climate model outputs is essential to correctly represent the distribution of precipitation, including extreme events which are often very localized, and missed by coarse-resolution models. Deep learning approaches have shown some skills to downscale precipitation at a much lower cost than physically-based models. However, it is still unclear how well these approaches capture extreme events with decadal return periods, which only occur a few times in the training dataset. We propose to study how a diffusion-based model downscales a particularly intense precipitation episode which took place in the South-East of France in October 2020, an event whose return period is estimated to be 100 years by Météo France [1].

2 Methods and data

Our diffusion model is based on the EDM framework from Karras et al [2] and is implemented using the IPSL-AID framework [6]. It was trained on a domain of approximate size 350 x 350 km^2 centered around Nice, France and uses precipitation and multiple covariates from the ERA5 dataset as predictors and precipitation from the regional CERRA reanalysis as predictand. The dataset consists of six-hourly fields spanning the years 1985 to 2014 for the training, and years 2015 to 2024 for the validation. An evaluation of the downscaling method using conventional ML metrics such as CRPS, power spectral density, and quantile-quantile plots shows that our model is capable of capturing typical precipitation. To further evaluate the ability of the model to downscale very rare events, we downscaled the situation of the 4 six-hourly timesteps between October 2nd 06:00 UTC to October 3rd 00:00 UTC ten times using the generative capability of our diffusion-based model. The maximum six-hour accumulated precipitation registered in the CERRA dataset during this episode is above 120

mm, much more than the 99.99 percentile of the training period, which is 36 mm. We then evaluated the downscaled precipitation using conventional metrics (CRPS, Fraction Skill Score, RAPSD) [3,4], PIT histograms [3], and checked the distribution of downscaled precipitation on a pixel basis as well as aggregated on hydrological basins. The diffusion model's performance is compared to two deterministic baselines: a UNet model and a classical statistical downscaling method (bias correction spatial disaggregation) [5].

3 Preliminary results and/or expected outcomes

The diffusion model is able to generate samples for moderate precipitation events that better capture the spatial structure (RAPSD) and distribution of precipitation than the considered baselines. When downscaling the considered extreme event, the diffusion model produces qualitatively satisfying, visually realistic samples that reproduce on average the location of the precipitation pattern observed, but underestimates or overestimates (sometimes substantially) the intensity depending on samples (see figure). The occurrence of rain at certain locations is also variable from sample to sample. This variability is also seen for precipitation aggregated on hydrographic basins (not shown). Because the event considered is exceptional in the studied dataset, it is difficult to know if this variability should be considered as an error of the downscaling model or if it is inherent to the distribution of fine scale precipitation. With the current design of the training procedure, the model is incapable of generating successive six-hourly samples that form a temporally coherent trajectory as samples are independent of their temporal predecessors. This leads to downscaled 12 hour- and 24 hour-accumulated precipitation fields that lack fine-scale details. We will investigate further this event and test ways to improve its simulation in a statistical sense.

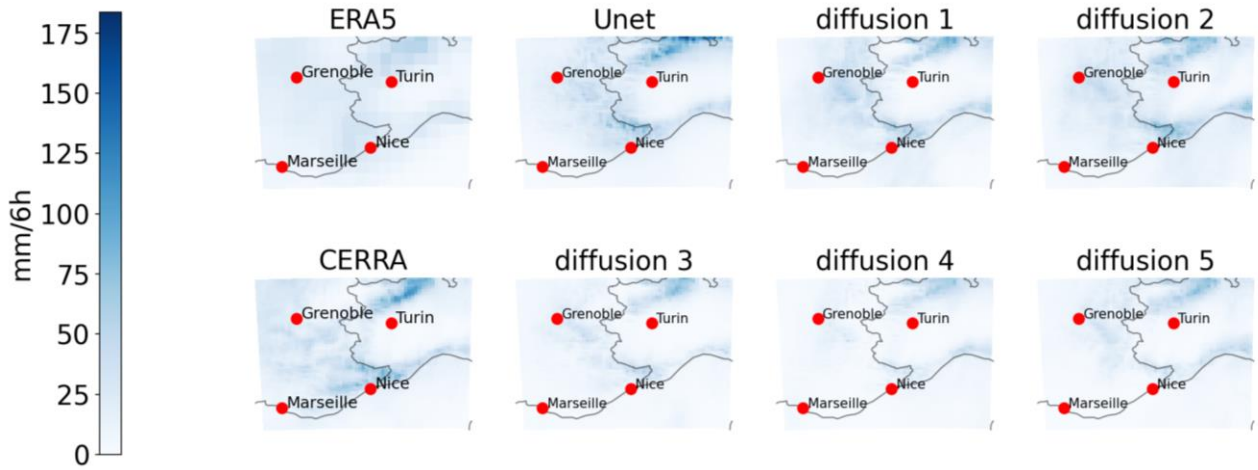

Figure 1. Example of six samples of downscaled six-hour accumulated precipitation fields (diffusion 0 to 5), compared to the corresponding ERA5 (low resolution) (top left) and CERRA (high resolution) (bottom left) precipitation for October 2nd, 12:00 to 18:00 UTC.

References

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